

For questions 1-3, Circle Q is shown, where chords $\overline{RC}$ and $\overline{WX}$ intersect at point P , $m\widehat{CW} = 142^\circ$, $m\widehat{RX} = 82^\circ$, and $m\angle WPC = (15x - 23)^\circ$.

1. Find the $m\angle WPC$.

$$\frac{142+82}{2} = 112^\circ$$

2. Write a formula to find the value of x .

$$15x - 23 = 112$$

3. Find the value of x .

$$15x - 23 = 112$$

$$15x = 135$$

$$x = 9$$

4. Circle E is shown, where chords $\overline{BD}$ and $\overline{CA}$ intersect at point Z . Find the value of x for $\angle AZD$, given $m\widehat{CD} = 115^\circ$ and $m\widehat{AB} = 53^\circ$.

$$\frac{115+53}{2} = 84$$

$$x = 180 - 84 = 96$$

For questions 5-7, Circle F is shown, where chords $\overline{LZ}$ and $\overline{KW}$ intersect at point G , $m\widehat{LW} = 70^\circ$, $m\widehat{KZ} = (15x + 10)^\circ$, and $m\angle KGZ = 145^\circ$.

5. Write a formula to find the value of x .

$$\frac{15x+10+70}{2} = 145$$

6. Find the value of x .

$$\frac{15x+10+70}{2} = 145$$

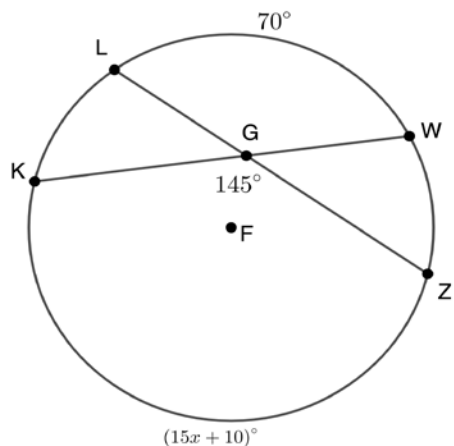
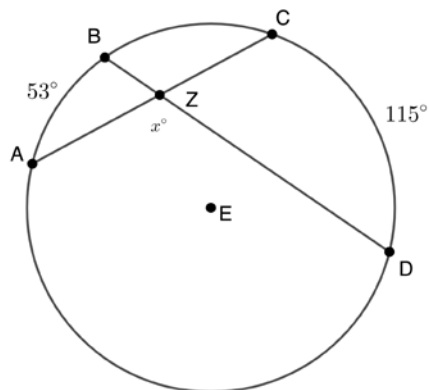
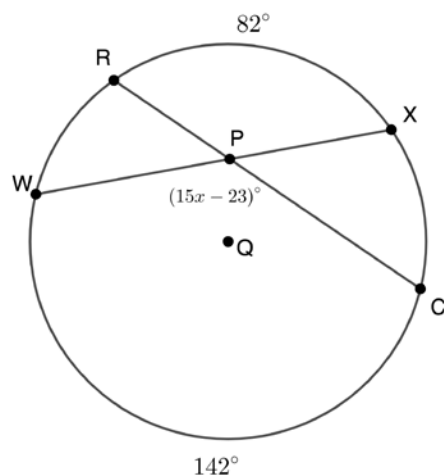
$$15x + 80 = 290$$

$$15x = 210$$

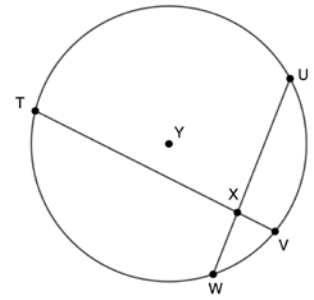
$$x = 14$$

7. Find the $m\widehat{KZ}$.

$$15(14) + 10 = 220^\circ$$



For questions 8-10, Circle Y is shown, where chords $\overline{VT}$ and $\overline{UW}$ intersect at point X .



8. If $m\widehat{TU} = 128^\circ$ and $m\widehat{VW} = 30^\circ$, what is the measure of $\angle TXU$?

$$\frac{128+30}{2} = 79^\circ$$

9. If $m\widehat{TW} = 152^\circ$ and $m\widehat{UV} = 40^\circ$, what is the measure of $\angle TXW$?

$$\frac{152+40}{2} = 96^\circ$$

10. If $m\widehat{VW} = 56^\circ$ and $m\angle TXU = 78^\circ$, what is the measure of $\widehat{TU}$?

$$78 = \frac{56+x}{2}$$

$$156 = 56 + x$$

$$100^\circ = x = \widehat{TU}$$